



JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA

Regulation: R23 | Academic Year 2023–24 Onwards

BASIC CIVIL & MECHANICAL ENGINEERING

B.Tech – I Year, I Semester

Common to All Engineering Branches

UNIT III – Transportation, Water Resources & Environmental Engineering

SUBJECT CODE

23BS1101 / 23BS1201

CREDITS

3 – 0 – 0

REGULATION

JNTUK R23

SEMESTER

I

UNIT

III of V

PAGES

20+ Pages

This module covers **Transportation Engineering, Water Resources Engineering, Hydrology, Rainwater Harvesting, Dams, Reservoirs** and **Environmental Engineering** with fully labeled SVG diagrams, comparison tables, formula boxes, and exam-focused Q&A.

Prepared for Academic / Examination Preparation Purposes | Based on JNTUK R23

Syllabus

Follows M.S. Palanisamy, S.S. Bhavikatti & Satheesh Gopi Reference Framework

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Syllabus Snapshot – Unit III

Topic Group	Topics Covered
Transportation Eng.	Importance, Highway Pavements (Flexible & Rigid), Harbour, Tunnel, Airport, Railway
Water Resources Eng.	Introduction, Sources, Quality, Specifications (IS 10500-2012)
Hydrology	Hydrological Cycle, Runoff, Infiltration, Rainwater Harvesting
Storage Structures	Dams – Types, Reservoirs – Functions

 **Exam Tip:** Unit III typically carries 14–20 marks in JNTUK exams. Focus on (i) Pavement comparison table, (ii) Rainwater harvesting diagram, (iii) Dam structure, and (iv) Water quality standards.

SECTION 1: INTRODUCTION TO TRANSPORTATION ENGINEERING

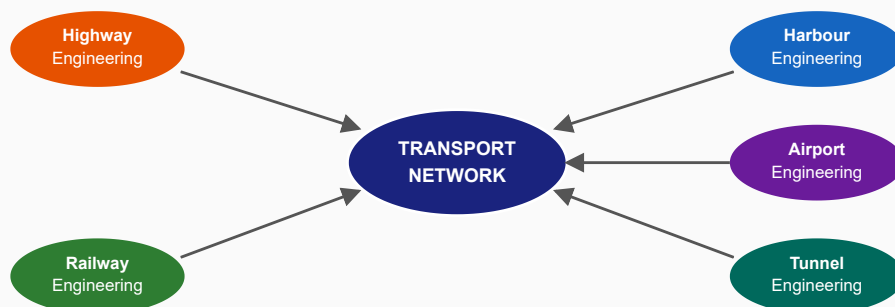
Definition: *Transportation Engineering is the branch of Civil Engineering that deals with the planning, design, operation, and management of safe and efficient movement of people and goods over land, water, and air using road, rail, harbour, tunnel, and airport infrastructure.*

Transportation forms the backbone of any developing nation. The field encompasses the design of geometric features of highways, pavement design, traffic control, railway track design, harbour construction, tunnel boring, and airport runway planning.

1.1 Scope of Transportation Engineering

- **Highway Engineering:** Design and construction of roads and pavements
- **Railway Engineering:** Track geometry, sleepers, ballast design
- **Airport Engineering:** Runway geometry, terminal design, pavement
- **Harbour Engineering:** Port layout, quays, jetties, breakwaters
- **Tunnel Engineering:** Underground passage construction through mountains/urban areas

Fig 1.1: Transportation Network Flow Diagram



SECTION 2: IMPORTANCE OF TRANSPORTATION IN NATIONAL ECONOMIC DEVELOPMENT

Transportation is the lifeline of a nation's economy. It connects production centers to consumption centers, enabling trade, industry, and social development. The development of a country can be gauged by the extent and quality of its transport infrastructure.

2.1 Role of Transportation in Economic Growth

- **Trade and Commerce:** Facilitates movement of raw materials, finished goods, and agricultural produce to markets, enabling economic activity.
- **Industrial Development:** Industries require efficient logistics for supply chain operations; good transport reduces production costs.
- **Agricultural Growth:** Roads in rural areas reduce post-harvest losses by connecting farms to markets quickly.
- **Employment Generation:** Construction, maintenance, and operation of transport systems create millions of direct and indirect jobs.
- **Regional Integration:** Connects backward regions with developed areas, reducing regional imbalances.
- **Tourism Promotion:** Good connectivity increases tourist influx, earning foreign exchange.
- **Defence and Strategic Value:** Roads and railways serve critical national defence requirements (e.g., border roads).
- **Social Development:** Improved access to education, healthcare, and social services in remote areas.
- **Urbanisation:** Supports planned urban growth and reduces congestion in cities through mass transit systems.

Key Fact (IS Standards): India has the **2nd largest road network** in the world (~63.7 lakh km). National Highways constitute only 2% of total road length but carry 40% of total road traffic.

KEY RELATION – TRANSPORT & DEVELOPMENT

Road Density = Total Road Length (km) / Area of Region (km²)

Freight Transport Efficiency = Tonne-km / Fuel consumed

Pavement Design Life (IRC) = Generally 15-20 years for NH, 10-15 years for SH

SECTION 3: TYPES OF HIGHWAY PAVEMENTS

Pavement: *A pavement is a durable hard surface layer constructed over the subgrade (natural ground) to provide a smooth, stable, and load-bearing surface for vehicular traffic.*

Highway pavements are broadly classified into two types based on their structural behavior and material composition:

- **1. Flexible Pavement** – Uses bituminous (asphalt) material; deflects under load
- **2. Rigid Pavement** – Uses cement concrete; does not deflect

3.1 Functions of Pavement

- Distributes wheel load to the subgrade safely
- Provides smooth riding surface for vehicles
- Resists abrasion due to traffic
- Prevents water from penetrating subgrade
- Provides adequate skid resistance

SECTION 4: FLEXIBLE PAVEMENTS

Definition: *Flexible pavement is a multi-layer pavement system made of bituminous materials that distributes vehicle loads to the subgrade through a series of granular and bound layers. It is called "flexible" because it deflects slightly under load and recovers after the load is removed.*

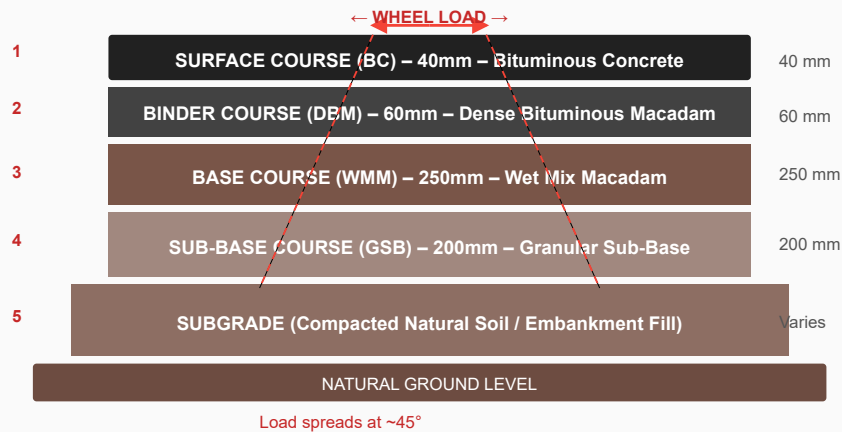
4.1 Layers of Flexible Pavement (Bottom to Top)

1. **Subgrade:** The natural soil/ground foundation. Compacted to bear all loads ultimately.
2. **Sub-base Course:** Granular material (gravel/WBM) layer that provides drainage and load spreading.
3. **Base Course:** Crushed stone/WMM (Wet Mix Macadam). Primary load-distributing layer.

4. **Binder Course (DBM):** Dense Bituminous Macadam – coarse aggregate + bitumen. Links base to surface.

5. **Surface Course (BC):** Bituminous Concrete – top wearing layer, smooth, skid-resistant.

Fig 4.1: Flexible Pavement – Layer Structure (Labeled Cross-Section)



4.2 Working Principle

Flexible pavement works on the principle of **load distribution through layers**. The wheel load is applied at the surface, and each successive layer distributes the load over a wider area so that the stress reaching the subgrade is within its bearing capacity. The load spreads approximately at 45° through each layer.

✓ Advantages

- Easy to construct and repair
- Lower initial cost
- Ready for traffic quickly
- Suitable for weak subgrade
- Can be strengthened progressively

✗ Disadvantages

- Higher maintenance cost
- Susceptible to rutting in hot weather
- Requires periodic renewal of surface
- Sensitive to water damage
- Lower design life (10–15 years)

🔑 Applications

Rural roads, state highways, low-traffic urban roads, airport taxiways, parking lots, pathways in residential colonies.

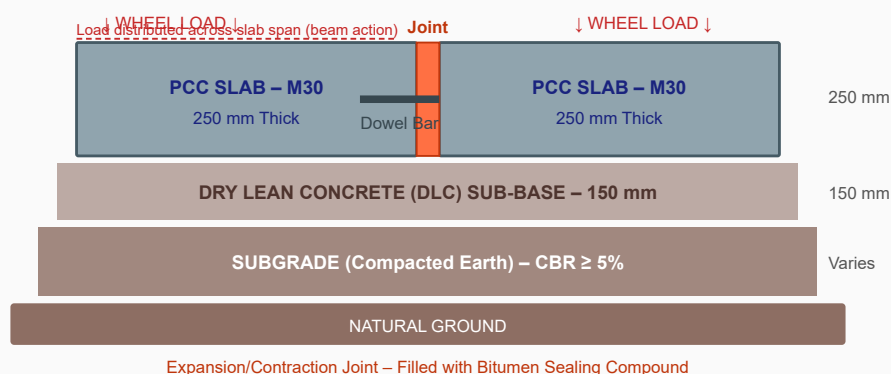
SECTION 5: RIGID PAVEMENTS

Definition: Rigid pavement is a pavement system constructed with Portland Cement Concrete (PCC) slabs that act as a structural slab to distribute wheel loads over a wide area of the subgrade. It is "rigid" because it does not deflect under load; instead it behaves like a beam/slab.

5.1 Layers of Rigid Pavement (Bottom to Top)

1. **Subgrade:** Natural/compacted earth base. Must be well compacted ($\text{CBR} \geq 5\%$).
2. **Sub-base (DLC):** Dry Lean Concrete – 100–150 mm layer providing uniform support and drainage.
3. **Concrete Slab (PCC):** M30/M40 grade cement concrete slab, 200–300 mm thick. Primary structural element.
4. **Joints:** Expansion, contraction, and construction joints to control cracking due to thermal expansion.
5. **Tie Bars & Dowel Bars:** Steel reinforcement at joints to transfer loads across slabs.

Fig 5.1: Rigid Pavement – Cross-Section Showing Layers & Joints



5.2 Working Principle

Rigid pavement works on **slab/beam action**. The concrete slab acts as a rigid structural element spanning over weak spots in the subgrade. It distributes the applied wheel load uniformly over a large area due to its high flexural strength. The load-distributing area is much larger compared to flexible pavement.

5.3 Types of Rigid Pavement

- **Jointed Plain Concrete Pavement (JPCP):** Plain concrete with joints at regular intervals
- **Jointed Reinforced Concrete Pavement (JRCP):** Lightly reinforced with wider joint spacing
- **Continuously Reinforced Concrete Pavement (CRCP):** No transverse joints; heavy steel reinforcement
- **Prestressed Concrete Pavement (PCP):** Prestressed tendons for longer slabs

▢ PAVEMENT DESIGN FORMULAS

Modulus of Elasticity of Concrete (E) = $5000 \times \sqrt{f_{ck}}$ (MPa) [IS 456]

Radius of Relative Stiffness (l) = $[E \cdot h^3 / 12(1-\mu^2) \cdot k]^{1/4}$

where h = slab thickness, k = modulus of subgrade reaction (MN/m³)

Equivalent Single Axle Load (ESAL) = (Actual Axle Load / Standard 80 kN)⁴

✓ Advantages

- Long design life (30–40 years)
- Low maintenance cost
- High load-bearing capacity
- Less susceptible to temperature
- Good visibility markings

✗ Disadvantages

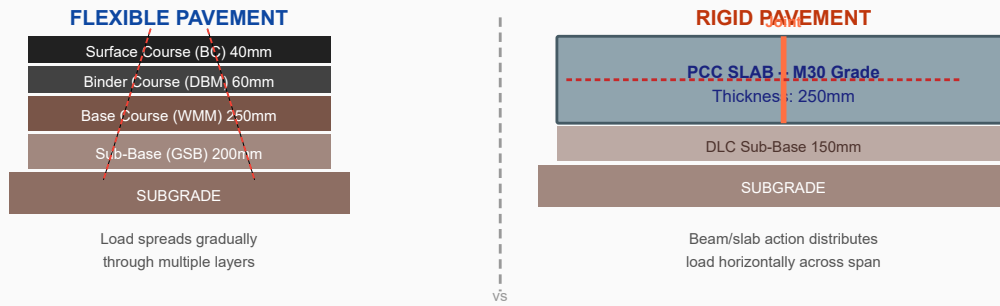
- High initial construction cost
- Requires curing period (28 days)
- Difficult to repair locally
- Joints cause discomfort
- Noise from joints

SECTION 6: COMPARISON OF FLEXIBLE AND RIGID PAVEMENTS

The following table provides a comprehensive comparison of flexible and rigid pavements on various parameters. **This is a very important exam topic** – memorize key points.

Sl. No.	Parameter	Flexible Pavement	Rigid Pavement
1	Material	Bitumen (Asphalt) + Aggregates	Portland Cement Concrete (PCC)
2	Structural Behavior	Deflects under load (flexible)	Does not deflect; acts as slab/beam
3	Load Transfer	By grain-to-grain contact; 45° spread	By slab action across wide area
4	Number of Layers	5 layers (subgrade to surface)	3 layers (subgrade + DLC + PCC slab)
5	Initial Cost	Low (₹1–2 crore/km for 2-lane)	High (₹2–4 crore/km for 2-lane)
6	Maintenance Cost	High (periodic overlay needed)	Low (minimal maintenance)
7	Design Life	10–15 years	30–40 years
8	Construction Time	Shorter (opens to traffic in 2–3 days)	Longer (28-day curing required)
9	Repair/Maintenance	Easy; localized patching possible	Difficult; slab must be replaced
10	Riding Quality	Smooth initially; roughens with time	Slightly rough due to joints
11	Temperature Effect	Rutting in summer; cracking in winter	Thermal cracking controlled by joints
12	Drainage Requirement	Good drainage essential	Less critical due to concrete impermeability
13	CBR of Subgrade	Works on weak subgrade (CBR 3–8%)	Requires good subgrade (CBR ≥ 5%)
14	Noise Level	Less noisy	More noisy at joints
15	Where Used	Rural roads, urban streets, parking	National Highways, expressways, airports

Fig 6.1: Side-by-Side Structural Comparison of Pavements



 **Exam Tip:** In 5-mark questions, draw the layer diagram and write at least 5 differences in a table format for maximum marks.

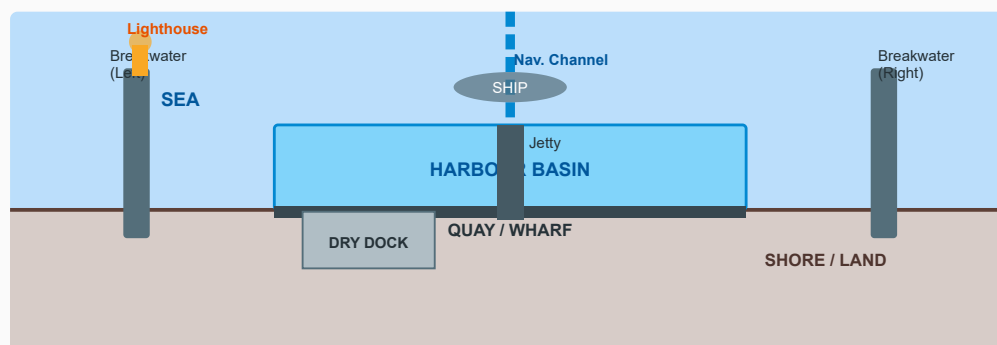
SECTION 7: HARBOUR ENGINEERING

Definition: *Harbour Engineering deals with the planning, design, and construction of harbours, ports, and associated marine structures to provide safe anchorage and loading/unloading facilities for ships and vessels.*

7.1 Key Components of a Harbour

- **Breakwaters:** Structures built to protect the harbour basin from wave action. Can be rubble mound or vertical wall type.
- **Quay / Wharf:** Loading/unloading platforms running parallel to the shore where ships berth.
- **Jetty / Pier:** Projecting structure perpendicular to shore allowing ship berthing on both sides.
- **Docks:** Enclosed water area where ships are loaded, repaired, or maintained.
- **Navigational Channel:** Dredged waterway for ship movement into the harbour.
- **Lighthouse:** Provides navigational guidance to ships approaching the harbour.

Fig 7.1: Plan View of Harbour Layout



SECTION 8: TUNNEL ENGINEERING

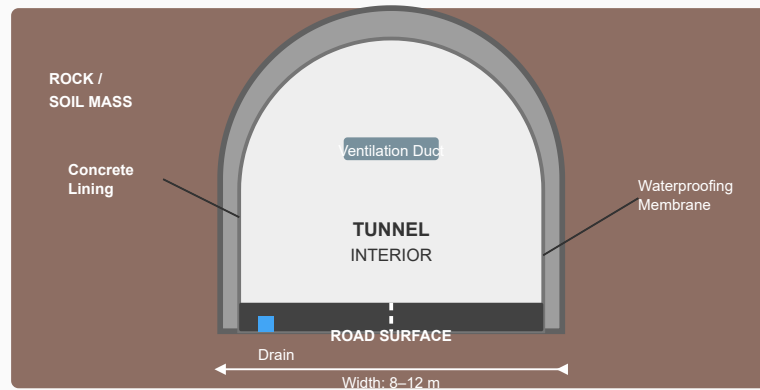
Definition: *Tunnel Engineering involves the design and construction of underground passages through mountains, under rivers, through urban areas, or under seabeds to carry roads, railways, water, utilities, or pedestrians.*

8.1 Types of Tunnels

- **Road Tunnels:** For vehicular traffic (e.g., Zoji La Tunnel, J&K)

- **Railway Tunnels:** For trains (e.g., Pir Panjal Tunnel)
- **Water Tunnels:** For conveying water for hydroelectric projects
- **Metro Tunnels:** For underground mass rapid transit systems

Fig 8.1: Tunnel Cross-Section (Horseshoe Shape).



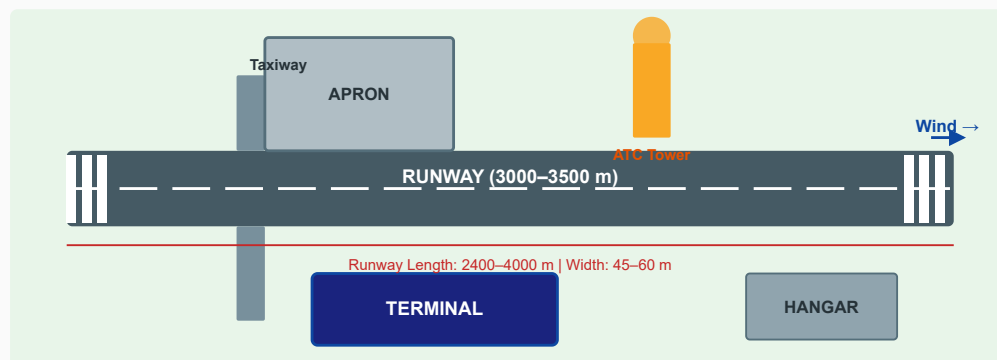
SECTION 9: AIRPORT ENGINEERING

Definition: *Airport Engineering deals with the planning, design, construction, and maintenance of airports, including runways, taxiways, aprons, terminal buildings, and supporting facilities for safe aircraft operations.*

9.1 Components of an Airport

- **Runway:** Paved strip for aircraft takeoff and landing. Oriented along the prevailing wind direction.
- **Taxiway:** Path connecting runway to apron/terminal, used by aircraft to taxi.
- **Apron:** Parking area for aircraft near terminal for boarding/deboarding passengers.
- **Terminal Building:** Passenger check-in, boarding, and customs facilities.
- **Hangar:** Maintenance and storage facility for aircraft.
- **Control Tower (ATC):** Air Traffic Control for managing takeoff/landing clearances.

Fig 9.1: Airport Runway & Taxiway Layout (Plan View)



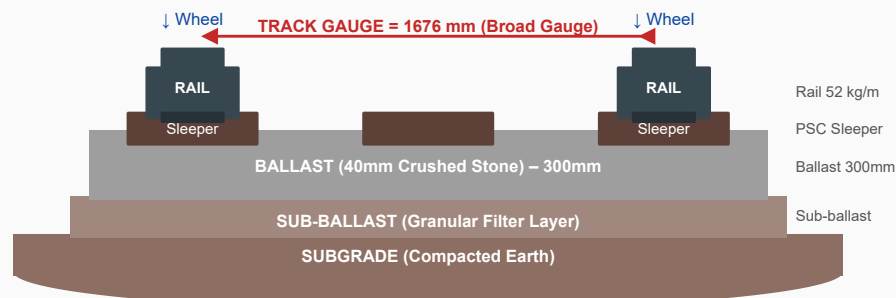
SECTION 10: RAILWAY ENGINEERING

Definition: *Railway Engineering is the branch of transportation engineering that deals with the design, construction, and maintenance of railways including tracks, stations, bridges, and signalling systems for train operations.*

10.1 Components of Railway Track Structure

- **Rails:** Steel sections (flat-bottomed T-rails); carry and guide the wheels. Standard weight: 52 kg/m in India.
- **Sleepers (Ties):** Transverse members supporting rails and maintaining track gauge. Types: Wooden, Concrete (PSC), Steel, Cast Iron.
- **Ballast:** Crushed stone (40mm size) bed providing drainage, distributing load, and preventing sleeper movement.
- **Sub-ballast:** Granular layer between ballast and subgrade acting as filter layer.
- **Subgrade:** Compacted earth foundation.
- **Track Gauge:** Distance between inner edges of rails. Broad Gauge (BG) = 1676 mm (India's main gauge).

Fig 10.1: Railway Track Structure – Cross-Section



SECTION 11: WATER RESOURCES ENGINEERING

Definition: *Water Resources Engineering is the branch of Civil Engineering concerned with the collection, storage, control, transportation, regulation, measurement, and use of water for the benefit of mankind.*

11.1 Introduction

Water is the most essential natural resource for life, agriculture, and industry. The Earth has a total water volume of about **1.39 billion km³**, but only 2.5% is freshwater, and of that, 70% is frozen in glaciers. Only 0.3% of all freshwater is accessible in rivers, lakes, and groundwater aquifers.

11.2 Sources of Water

Water for human use can be obtained from the following sources:

Source Type	Examples	Characteristics
Surface Water	Rivers, lakes, ponds, tanks, streams	Readily available; may be turbid; seasonal variation
Ground Water	Wells (open & tube wells), springs, artesian wells	Filtered by soil; less turbid; more minerals
Rainwater	Rooftop harvesting, tanks	Purest natural form; requires collection system
Sea Water	After desalination	Available in coastal areas; treatment required

11.3 Quality of Water

Quality of water is assessed based on three parameters:

- Physical Parameters:** Colour, turbidity, taste, odour, temperature
- Chemical Parameters:** pH, hardness, dissolved salts (TDS), dissolved oxygen
- Biological Parameters:** Bacteria count (E. coli), algae, viruses

11.4 Water Quality Standards – IS 10500: 2012 (Drinking Water)

Parameter	Acceptable Limit	Permissible Limit (if no alt.)	Unit
pH Value	6.5 – 8.5	No relaxation	–

Turbidity	1 NTU	5 NTU	NTU
Total Dissolved Solids	500	2000	mg/L
Total Hardness	200	600	mg/L as CaCO ₃
Chlorides	250	1000	mg/L
Sulphates	200	400	mg/L
Iron	0.3	1.0	mg/L
Arsenic	0.01	0.05	mg/L
Nitrates	45	45	mg/L
Fluorides	1.0	1.5	mg/L
E.coli	Absent	Absent	per 100 mL
Dissolved Oxygen (DO)	≥ 4	≥ 4	mg/L

📌 **IS Reference:** IS 10500:2012 – "Indian Standard Drinking Water Specification (Second Revision)" by Bureau of Indian Standards is the key reference for water quality parameters in India.

📐 IMPORTANT WATER QUALITY FORMULAS

BOD (Biochemical Oxygen Demand) = $DO_0 - DO_t$ [mg/L]

COD (Chemical Oxygen Demand) = Oxygen required to oxidize all organic matter

Hardness (as CaCO₃) = $[Ca^{2+} \times 2.5] + [Mg^{2+} \times 4.1]$ mg/L

Turbidity Removal Efficiency = $(1 - C_{out}/C_{in}) \times 100 \%$

Flow Rate: $Q = A \times V$ [m³/s] where A = cross-section area, V = velocity

11.5 Water Treatment Process

Step-by-Step Water Treatment:

Step 1 – Screening: Removal of large floating debris using bar screens at intake

Step 2 – Coagulation: Addition of alum [Al₂(SO₄)₃] to form flocs; neutralizes colloidal particles

Step 3 – Flocculation: Gentle mixing to aggregate flocs into settleable masses

Step 4 – Sedimentation: Flocs settle in sedimentation tank (2–8 hours residence time)

Step 5 – Filtration: Rapid Sand Filter – removes remaining fine particles and microorganisms

Step 6 – Disinfection: Chlorination (0.2–0.5 mg/L residual chlorine) to kill bacteria

Step 7 – pH Correction: Lime dosing to maintain pH 6.5–8.5

Step 8 – Distribution: Treated water pumped to overhead tanks and distribution mains

SECTION 12: HYDROLOGY & RAINWATER HARVESTING

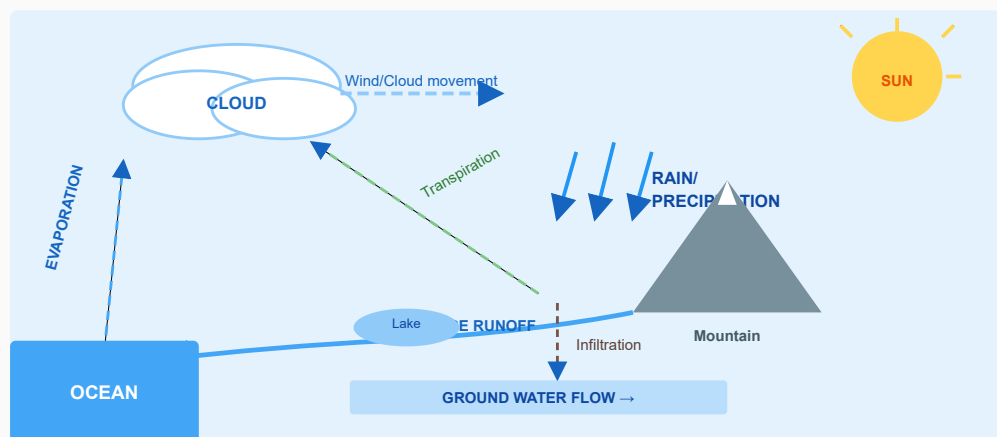
12.1 Introduction to Hydrology

Definition: *Hydrology is the science that deals with the occurrence, distribution, movement, and properties of water on the Earth – encompassing the atmosphere, surface, and subsurface, including the hydrological cycle (water cycle).*

12.2 The Hydrological Cycle

The hydrological cycle (water cycle) is the continuous process by which water evaporates from oceans, condenses as clouds, precipitates as rain/snow, flows as runoff, and returns to the ocean.

Fig 12.1: Hydrological (Water) Cycle – Schematic Diagram



KEY HYDROLOGICAL EQUATIONS

Water Balance Equation: $P = R + ET + \Delta S$

where P = Precipitation, R = Runoff, ET = Evapotranspiration, ΔS = Change in storage

Rational Formula (Peak Runoff): $Q = C \times i \times A / 360 \text{ [m}^3/\text{s]}$

where C = Runoff coefficient, i = Rainfall intensity (mm/hr), A = Catchment area (ha)

Runoff = Rainfall - Losses (Infiltration + Evaporation + Depression storage)

Rainfall Intensity (i) = $a / (t_c + b)$ [Talbot's formula] where t_c = time of concentration

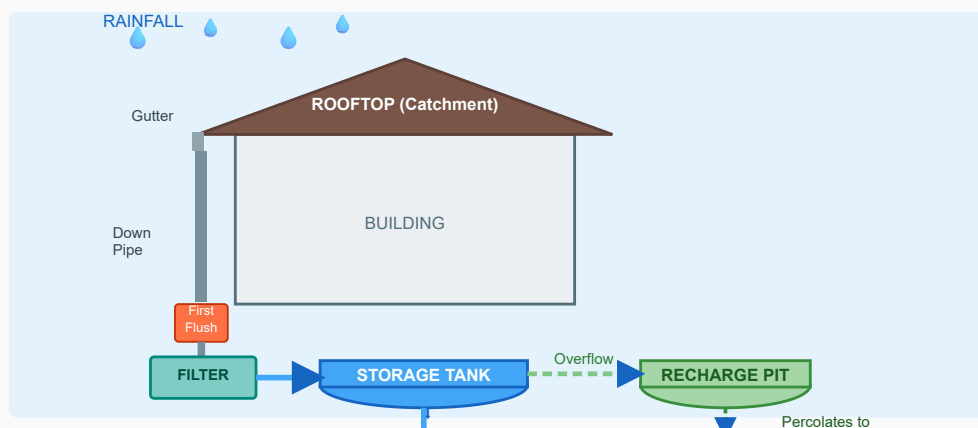
12.3 Rainwater Harvesting

Definition: *Rainwater Harvesting (RWH) is the process of collecting, filtering, storing, and utilizing rainwater that falls on rooftops or land surfaces for domestic, agricultural, or groundwater recharge purposes.*

12.4 Components of Rooftop Rainwater Harvesting System

1. **Collection Surface (Rooftop):** Roof of building (RCC, GI sheet) acts as catchment area
2. **Gutters/Drains:** Channels at roof edge to collect rainwater from rooftop
3. **Downpipes:** Vertical pipes carrying water from gutters to storage/filtration
4. **First Flush Device:** Diverts first 10–20 minutes of rain (which is dirty) away; clean subsequent rain is collected
5. **Filter Unit:** Multi-layer filter (sand, gravel, charcoal) to remove suspended particles
6. **Storage Tank:** Underground/overhead sump for storing filtered rainwater
7. **Recharge Pit/Well:** Allows excess water to percolate underground and recharge aquifer

Fig 12.2: Rooftop Rainwater Harvesting System



✓ Benefits of Rainwater Harvesting

- Reduces water bills and dependency on municipal supply
- Recharges depleting groundwater table
- Reduces urban flooding and stormwater runoff
- Reduces soil erosion
- Provides water during drought conditions

✗ Limitations

- Rainfall dependent – unreliable in dry areas
- Initial cost of system installation
- Requires regular maintenance of filters
- Risk of contamination if not properly managed
- Limited storage volume for large demands

RAINWATER HARVESTING POTENTIAL CALCULATION

Annual Harvestable Volume = Roof Area (m^2) $\times$ Annual Rainfall (m) $\times$
Runoff Coefficient (C)

where C = 0.75-0.90 for concrete/tile roofs (IS 15797)

Example: 100 m^2 roof, 800 mm rain, C = 0.8 $\rightarrow$ Volume = $100 \times 0.8 \times 0.8 =$
64 m^3/year

SECTION 13: WATER STORAGE & CONVEYANCE

STRUCTURES – DAMS AND RESERVOIRS

13.1 Dams

Definition: A dam is a hydraulic structure built across a river, stream, or watercourse to impound water, create a reservoir for storage, control floods, or divert water for irrigation, water supply, and hydroelectric power generation.

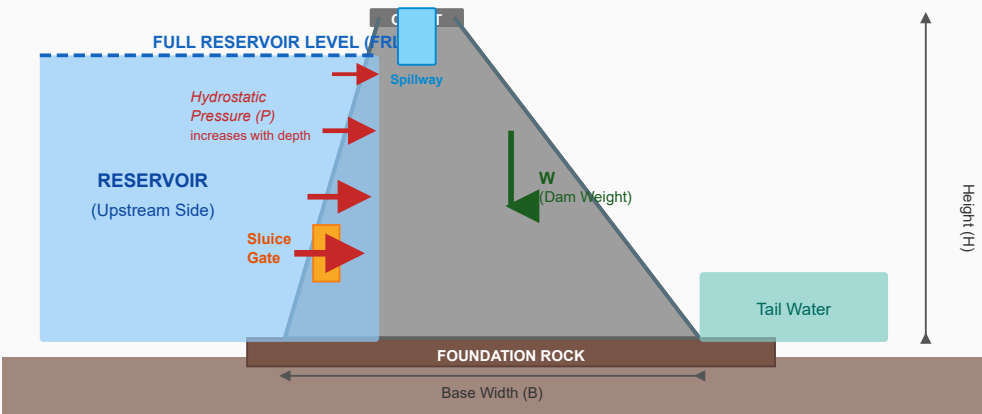
13.2 Classification of Dams

Basis	Types	Description
Material	Earthen Dam, Gravity Dam, Arch Dam, Rockfill Dam, Masonry Dam	Based on material of construction
Hydraulic Design	Storage Dam, Diversion Dam, Detention Dam	Based on primary function
Structure	Gravity Dam, Arch Dam, Buttress Dam, Embankment Dam	Based on structural behavior

13.3 Gravity Dam – Detailed Study

A gravity dam resists horizontal water pressure (hydrostatic force) through its own weight. The section is a triangle with a broad base. Made of concrete or masonry.

Fig 13.1: Gravity Dam Cross-Section with Labels



DAM DESIGN FORMULAS

Hydrostatic Pressure: $P = (1/2) \times \gamma_w \times H^2$ [kN/m run]

where $\gamma_w = 9.81$ kN/m³, H = water depth (m)

Factor of Safety against sliding: $FOS = \mu \times W / P \geq 1.5$

Factor of Safety against overturning: $FOS = (\text{Stabilizing moment}) / (\text{Overturning moment}) \geq 1.5$

Reservoir Storage: $S = \text{Inflow Volume} - \text{Outflow Volume} - \text{Evaporation Losses}$

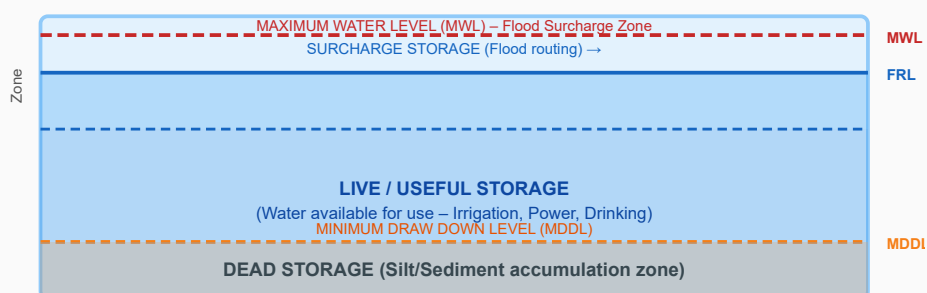
13.4 Reservoirs

Definition: A reservoir is a large natural or artificial storage body of water formed behind a dam, used to store water for irrigation, drinking water supply, flood control, and hydroelectric power generation.

13.5 Types of Reservoirs

- **Storage/Conservation Reservoir:** Stores water during flood season for use in dry season
- **Distribution Reservoir:** Small service reservoir in a water supply distribution system
- **Flood Control Reservoir:** Retains floodwaters to prevent downstream flooding
- **Multi-purpose Reservoir:** Serves irrigation, power, water supply, and recreation simultaneously

Fig 13.2: Reservoir Storage Zones



SECTION 14: ENVIRONMENTAL ENGINEERING – INTRODUCTION AND IMPORTANCE

Definition: *Environmental Engineering is the branch of Civil Engineering that applies scientific and engineering principles to protect the environment from the harmful effects of human activities, improve environmental quality, and ensure sustainable use of natural resources. It deals with water supply, sewage treatment, solid waste management, air pollution control, and noise abatement.*

14.1 Importance of Environmental Engineering

- **Safe Drinking Water:** Provides treated, potable water to communities, preventing waterborne diseases.
- **Wastewater Treatment:** Treats sewage and industrial effluents before discharge to water bodies.
- **Solid Waste Management:** Scientific disposal of municipal and hazardous solid wastes.
- **Air Pollution Control:** Design of emission control systems for industries and vehicles.
- **Environmental Impact Assessment (EIA):** Evaluates impacts of development projects before approval.
- **Public Health Protection:** Prevention of epidemics through sanitation and hygiene infrastructure.
- **Biodiversity Conservation:** Protection of ecosystems from industrial and developmental pressures.
- **Climate Change Mitigation:** Designing green infrastructure and reducing carbon footprint.

14.2 Scope of Environmental Engineering

Sub-field	Coverage
Water Supply Engineering	Source development, water treatment, distribution network design
Sewerage Engineering	Sewer design, sewage treatment plants (STP), disposal systems
Solid Waste Engineering	Collection, segregation, composting, landfill design
Air Pollution Control	Stack emission monitoring, ESP, scrubbers, green belts

Noise Pollution	Noise barriers, soundproofing, land use planning
Industrial Waste Management	Treatment of heavy metals, chemicals in industrial effluents
Environmental Impact Assessment	Pre-project study for roads, dams, industries under EIA Notification 2006

14.3 Types of Pollution and Their Impacts

Type of Pollution	Sources	Effects	Remediation
Water Pollution	Industrial effluents, agricultural runoff, sewage	Eutrophication, disease, fish kill	STP, ETP, constructed wetlands
Air Pollution	Industries, vehicles, power plants	Respiratory disease, acid rain, global warming	ESPs, scrubbers, catalytic converters
Soil Pollution	Pesticides, heavy metals, solid waste dumping	Reduced fertility, food chain contamination	Bioremediation, phytoremediation
Noise Pollution	Traffic, industries, aircraft	Hearing loss, stress, hypertension	Noise barriers, green belts

14.4 Key Environmental Standards (India)

Parameter	Standard/Act	Limit / Requirement
Drinking Water	IS 10500:2012	pH 6.5–8.5, TDS $\leq$ 500 mg/L
Ambient Air Quality	NAAQS 2009	PM _{2.5} $\leq$ 40 $\mu\text{g}/\text{m}^3$ (annual)
Industrial Effluents	General Standards – Env. Protection Act 1986	pH 6.5–9.0, BOD $\leq$ 30 mg/L
Noise (Day/Night)	Noise Pollution Rules 2000	Residential: 55/45 dB(A)



Key Environmental Laws in India:

- Environment Protection Act, 1986 – Umbrella act for environmental protection
- Water (Prevention and Control of Pollution) Act, 1974
- Air (Prevention and Control of Pollution) Act, 1981
- EIA Notification, 2006 – Mandatory EIA for major projects
- National Green Tribunal (NGT) Act, 2010 – Fast-track environmental dispute resolution

SECTION 15: QUESTIONS & ANSWERS

15.1 Two-Mark Questions & Answers

2 Marks

Q1. Define flexible pavement. State two advantages.

Flexible Pavement: A flexible pavement is a multi-layer pavement constructed using bituminous (asphalt) materials that distributes wheel loads to the subgrade through granular and bound layers. It is called "flexible" because it deflects slightly under load and recovers after load removal.

Advantages:

- (i) Easy and economical to construct with lower initial cost compared to rigid pavement.
- (ii) Can be opened to traffic very quickly (within 2–3 days after construction).
- (iii) Can be strengthened progressively by adding overlay courses.
- (iv) Suitable for weak subgrade soils where CBR values are low.

2 Marks

Q2. What is a harbour? Name its main components.

Harbour: A harbour is a sheltered body of water along a coast where ships can anchor or dock safely for loading/unloading passengers and cargo, and for shelter from storms and adverse weather conditions. It is a vital node in maritime transportation.

Main Components of a Harbour:

- (i) **Breakwater** – Protects harbour basin from wave action
- (ii) **Quay/Wharf** – Loading and unloading platform parallel to shore
- (iii) **Jetty/Pier** – Projecting structure for ship berthing
- (iv) **Dock** – Enclosed space for ship repair and maintenance
- (v) **Navigational Channel** – Dredged path for ship access
- (vi) **Lighthouse** – Navigational guide for approaching vessels

2 Marks

Q3. What is the water balance equation? State all terms.

Water Balance Equation: The hydrological water balance equation relates all inflows and outflows for a catchment over a given time period:

$$P = R + ET + \Delta S$$

Where:

P = Precipitation (rainfall/snowfall in mm or m³)

R = Runoff (surface flow reaching rivers/streams in mm or m³)

ET = Evapotranspiration (evaporation from water bodies + transpiration from plants)

ΔS = Change in groundwater and soil moisture storage

This equation expresses the conservation of mass principle for the hydrological system. When $\Delta S = 0$ (steady state), $P = R + ET$.

2 Marks

Q4. What is a dam? Name its types.

Dam: A dam is a hydraulic structure constructed across a river or stream to store water, control floods, divert water for irrigation, provide drinking water supply, and generate hydroelectric power. The stored water body behind the dam is called a reservoir.

Types of Dams:

Based on Material: (i) Earthen/Embankment Dam, (ii) Gravity Dam (Concrete/Masonry), (iii) Arch Dam, (iv) Rockfill Dam

Based on Function: (i) Storage Dam, (ii) Detention/Flood Control Dam, (iii) Diversion Dam

Based on Structure: (i) Gravity Dam (resists by self-weight), (ii) Arch Dam (transfers load to abutments), (iii) Buttress Dam (with counterforts for support)

Notable Indian examples: Bhakra Nangal (Gravity), Hirakud (Earthen), Idukki (Arch), Tehri (Rockfill).

2 Marks

Q5. State the IS 10500:2012 limits for pH and TDS of drinking water.

IS 10500:2012 – Indian Standard for Drinking Water Specifications:

1. pH Value:

Acceptable Limit: **6.5 to 8.5**

Permissible Limit: No relaxation permitted. Water outside this range must not be supplied.

Significance: Below 6.5 is acidic (corrosive to pipes); above 8.5 is alkaline (causes scaling).

2. Total Dissolved Solids (TDS):

Acceptable Limit: **500 mg/L**

Permissible Limit (if no alternate source): **2000 mg/L**

Significance: High TDS causes unpleasant taste, pipe scaling, and health hazards if metals are present. TDS includes calcium, magnesium, sodium, potassium salts dissolved in water.

2 Marks

Q6. Define hydrology. What is the hydrological cycle?

Hydrology: Hydrology is the science that deals with the occurrence, distribution, and movement of water on the surface, subsurface, and in the atmosphere of the Earth. It studies precipitation, runoff, evapotranspiration, infiltration, and groundwater flow.

Hydrological Cycle (Water Cycle): The hydrological cycle is the continuous, closed-loop process by which water moves between the ocean, atmosphere, land surface, and subsurface. The steps are:

- (i) Evaporation from oceans and water bodies by solar energy
- (ii) Condensation of water vapor to form clouds
- (iii) Precipitation as rain, snow, or hail over land and sea
- (iv) Interception by vegetation and surface storage
- (v) Infiltration into soil and percolation to groundwater
- (vi) Surface runoff flowing to rivers and eventually back to the ocean

The water balance equation governing this: $P = R + ET + \Delta S$

15.2 Five-Mark Questions & Answers

5 Marks

Q7. Explain flexible pavement with a neat labeled diagram. Compare it with rigid pavement. (5 marks)

FLEXIBLE PAVEMENT:

Definition: Flexible pavement is a multi-layer pavement system made using bituminous materials. It distributes the wheel load to the subgrade through successive layers. The term "flexible" indicates that the pavement deflects slightly under wheel load and recovers after load is removed, similar to a flexible beam.

Layer Structure (from bottom to top):

- 1. Subgrade:** Compacted natural ground/embankment fill. Ultimate load bearer. CBR test determines its strength.
- 2. Granular Sub-base (GSB):** 200 mm granular layer (gravel + sand). Provides drainage, prevents frost heave, and distributes load over subgrade.
- 3. Wet Mix Macadam (WMM) Base Course:** 250 mm crushed stone mixed with water and compacted. Primary load-distributing layer.
- 4. Dense Bituminous Macadam (DBM) Binder Course:** 60 mm. Coarse aggregate + bitumen binder. Connects base and surface course. Provides structural strength.
- 5. Bituminous Concrete (BC) Surface Course:** 40 mm top wearing layer. Smooth, waterproof, and skid-resistant surface.

Working Principle: The wheel load applied at the surface is distributed through successive layers at approximately 45° spread angle. Each layer receives a lower stress than the one above it, so the subgrade receives only a small safe stress. This is called "load spreading" or "stress attenuation."

Pavement Design Life: 15 years for National Highways (IRC:37-2018)

Key Comparison:

- Flexible pavement uses bitumen; Rigid uses cement concrete
- Flexible deflects; Rigid does not deflect (beam action)
- Flexible has higher maintenance; Rigid has lower maintenance
- Flexible has lower initial cost; Rigid has lower life-cycle cost
- Flexible has 10–15 year life; Rigid has 30–40 year life

(Refer to Fig 4.1 on Page 4 for labeled diagram of flexible pavement cross-section)

5 Marks

Q8. Explain rainwater harvesting with a neat labeled diagram. (5 marks)

RAINWATER HARVESTING (RWH):

Definition: Rainwater Harvesting is the process of collecting, storing, and utilizing rainwater that falls on rooftops or land surfaces, for purposes such as domestic use, irrigation, and recharging groundwater aquifers.

Need for Rainwater Harvesting:

- (i) Growing water scarcity and depleting groundwater levels
- (ii) Increasing urban flooding due to impervious surfaces
- (iii) Dependency on monsoon rainfall in India (average 1170 mm/year)
- (iv) To supplement municipal water supply

Components and Working:

- 1. Catchment Area (Rooftop):** The roof surface (RCC/tile/metal sheet) collects rainwater. Cleanliness of roof affects water quality.
- 2. Gutters:** Channels along the roof edge that collect runoff from the roof surface and direct it to downpipes.
- 3. Downpipes:** Vertical PVC/GI pipes carrying water from gutters to ground level.
- 4. First Flush Device:** A diverter that discards the first 15–20 minutes of rainfall (which contains dirt, bird droppings, dust) and directs cleaner subsequent rainfall to storage. $\text{Volume diverted} = 2.5 \text{ mm depth} \times \text{roof area}$.
- 5. Filter Unit:** Multi-layer filter consisting of: Top – Coarse gravel; Middle – Fine sand; Bottom – Activated charcoal/stone. Removes suspended particles, bacteria, and odour.
- 6. Storage Tank:** RCC underground sump or overhead tank to store filtered water. Sized based on daily demand and number of dry days.
- 7. Recharge Pit/Bore:** Excess water that cannot be stored is directed to a recharge pit ($3\text{m} \times 3\text{m} \times 3\text{m}$ gravel-filled pit) or recharge bore to replenish the groundwater aquifer.

Harvesting Potential Formula:

$\text{Volume} = \text{Roof Area (m}^2\text{)} \times \text{Rainfall (m)} \times \text{Runoff Coefficient (C = 0.8)}$

Benefits: Reduces water bills, recharges groundwater, reduces urban flooding, reduces soil erosion.

(Refer to Fig 12.2 on Page 10 for labeled diagram of RWH system)

5 Marks

Q9. Explain the structure of a gravity dam with a neat sketch and important design considerations. (5 marks)

GRAVITY DAM:

Definition: A gravity dam is a solid concrete or masonry dam that resists the horizontal water pressure (hydrostatic force) and other forces by its own dead weight. The cross-section is typically triangular or trapezoidal, broad at the base and narrow at the top (crest).

Forces Acting on a Gravity Dam:

- (i) **Water Pressure (P):** Horizontal force = $\frac{1}{2} \times \gamma_w \times H^2$ (increases with depth, acting on upstream face)
- (ii) **Self-Weight (W):** Vertical downward force = Volume $\times$ Unit weight of concrete (24 kN/m³)
- (iii) **Uplift Pressure (U):** Water seeping under foundation creates upward pressure; reduced by drainage curtain
- (iv) **Silt/Sediment Pressure:** Horizontal pressure from silt deposited on upstream face
- (v) **Earthquake Force:** For dams in seismic zones, pseudo-static horizontal force added
- (vi) **Wave Pressure:** Impact of waves on upstream face

Design Criteria (Safety Requirements):

- (a) FOS against sliding ≥ 1.5 ($\mu W/P$, $\mu = 0.6\text{--}0.75$ for concrete on rock)
- (b) FOS against overturning ≥ 1.5
- (c) No tension at any cross-section (middle-third rule: $e \leq B/6$)
- (d) Compressive stress $\leq$ Permissible stress of material

Key Structural Features:

- Upstream face: Slightly inclined or vertical
- Downstream face: Sloped at 0.7–0.8 horizontal:1 vertical
- Crest width: $1/14 \times$ dam height (minimum for road) or 1–3 m for walkway
- Base width: 60–70% of dam height for gravity action
- Spillway: Provided at crest to discharge excess water during floods
- Sluice gates: Provided at lower levels for silt flushing and controlled release
- Cutoff wall: Extends into foundation rock to prevent seepage

Examples: Bhakra Dam (Himachal Pradesh, 226 m high gravity dam), Mettur Dam (Tamil Nadu)

(Refer to Fig 13.1 on Page 11 for labeled gravity dam cross-section)

5 Marks

Q10. Write about the importance of transportation engineering in India's economic development. (5 marks)

IMPORTANCE OF TRANSPORTATION IN ECONOMIC DEVELOPMENT:

Transportation is often called the "engine of economic growth." No modern economy can function without efficient movement of people, goods, and services. In India, transportation plays a particularly vital role due to its large geography, diverse population, and growing economy.

1. Trade and Commerce (GDP Contribution): Transport links production zones to markets. Road and rail freight account for 85% of India's freight movement. Without efficient transport, goods cannot reach consumers, leading to market failures and price rise.

2. Agricultural Sector: India's agriculture contributes ~18% to GDP. Efficient road connectivity reduces post-harvest losses (currently 20–30% due to poor connectivity) by enabling rapid farm-to-market movement. PM Gram Sadak Yojana linked over 1.7 lakh villages to road networks.

3. Industrial Development: Industries require raw material supply and finished goods distribution. Industrial corridors (Delhi-Mumbai, Chennai-Bengaluru) rely heavily on multimodal transportation (road + rail + port).

4. Employment Generation: The construction, maintenance, and operation of transportation infrastructure provides 50 million direct and indirect jobs in India (road construction alone employs ~4 crore workers).

5. Regional Equity and Social Integration: Good transport connects backward regions with growth centers. Northeast connectivity projects (Bogibeel Bridge, Atal Tunnel) are transforming regional economies and ensuring national integration.

6. Defence and National Security: Border roads (BRO – Border Roads Organisation) are critical for troop movement and supply. The Manali-Leh highway and Atal Tunnel are strategic examples.

7. Tourism and Export Earnings: Airports and ports drive tourism. India's 14 major ports handled 795 MT of cargo in 2022-23, earning significant foreign exchange.

8. Urban Development: Metro rail and expressways are essential for managing urbanization. Mumbai Metro, Delhi Metro, Hyderabad Metro serve millions daily, reducing road congestion.

Conclusion: Every ₹1 invested in transport infrastructure generates ₹2–3 in GDP growth (economic multiplier effect). India's National Infrastructure Pipeline (NIP) targets ₹111 lakh crore in infrastructure investment by 2025, with transport being the largest component.

SECTION 16: QUICK REVISION SHEET – UNIT III

⚡ One-page summary of all critical points for last-minute exam preparation

Pavement Key Points

- **Flexible:** Bitumen, 5 layers, 10–15 yr life, deflects
- **Rigid:** PCC M30, 3 layers, 30–40 yr life, beam action
- Flexible = Low initial cost; High maintenance
- Rigid = High initial cost; Low maintenance
- Layers (Flexible): SC → BC → WMM → GSB → Subgrade
- CBR test – strength of subgrade

Water Quality (IS 10500:2012)

- pH: 6.5 – 8.5 (No relaxation)
- TDS: 500 mg/L (2000 permissible)
- Turbidity: 1 NTU (5 permissible)
- Hardness: 200 mg/L (600 permissible)
- Chlorides: 250 mg/L (1000 permissible)
- Fluorides: 1.0 mg/L (1.5 permissible)
- E.coli: Absent (no relaxation)

Key Formulas

- $Q = A \times V$ (Flow rate)
- $P = R + ET + \Delta S$ (Water balance)
- $Q = C \cdot i \cdot A / 360$ (Rational formula)
- $P = \frac{1}{2} \gamma_w \cdot H^2$ (Hydrostatic force on dam)
- Harvestable Vol = Area $\times$ Rain $\times$ C
- Runoff = Rainfall – Losses
- FOS slide = $\mu W / P \geq 1.5$

Hydrology Terms

- Evaporation – water → water vapour
- Transpiration – plant water release
- Precipitation – rain/snow/hail
- Infiltration – water into soil
- Runoff – surface flow to rivers
- Groundwater – below water table
- Water Table – top of saturated zone

Dam Quick Facts

- Gravity Dam – resists by own weight
- Arch Dam – transfers to abutments
- Earthen Dam – cheapest; uses local soil
- Reservoir zones: MWL → FRL → MDDL → Dead
- Live storage – usable zone
- Dead storage – silt accumulation
- Bhakra = Gravity | Idukki = Arch

Transport Basics

- Track gauge (BG) = 1676 mm
- Rail weight = 52 kg/m (standard India)
- Sleeper types: Wood, PSC, Steel
- Ballast – 40 mm crushed stone, 300 mm
- Runway length = 2400–4000 m
- Harbour: Breakwater protects basin
- Tunnel shape: Horseshoe (road/rail)

Environmental Eng.

- BOD = $DO_0 - DOT$ (oxygen demand)

Rainwater Harvesting

- C (RCC roof) = 0.80 – 0.90

- COD = chemical oxygen demand
- STP = Sewage Treatment Plant
- ETP = Effluent Treatment Plant
- EIA = Environmental Impact Assessment
- Ambient PM_{2.5} ≤ 40 µg/m³ (NAAQS)
- NGT Act 2010 – green tribunal

- First flush = first 10–20 min diverted
- Filter: Gravel → Sand → Charcoal
- Mandatory: Chennai, Bengaluru, Delhi
- Recharge pit – replenishes groundwater
- IS 15797 – RWH guidelines
- India avg rainfall = 1170 mm/year

Pavement Comparison – Quick Table

Parameter	Flexible	Rigid
Material	Bitumen + aggregates	Cement concrete (M30)
Behavior	Deflects (flexible)	No deflection (beam action)
Layers	5 (SC, BC, WMM, GSB, Subgrade)	3 (PCC, DLC, Subgrade)
Initial Cost	Low	High
Maintenance	High	Low
Design Life	10–15 years	30–40 years
Construction Time	Short (2–3 days to open)	Long (28-day curing)
Joint	Not required	Required (expansion/contraction)

Top 5 Exam-Guaranteed Topics from Unit III:

1. Flexible pavement – layers + diagram + advantages ✓
2. Rainwater harvesting – components + diagram ✓
3. Gravity dam – cross-section + forces + FOS formulas ✓
4. Rigid vs Flexible comparison table ✓
5. IS 10500:2012 water quality parameters ✓

ADDITIONAL DETAIL: TRANSPORT SYSTEMS – KEY TECHNICAL POINTS

A. Airport Engineering – Technical Details

A.1 Geometric Design of Runways

Parameter	Value / Standard
Runway Length (domestic)	1500–2400 m
Runway Length (international)	3000–4000 m
Runway Width	45 m (Code C/D), 60 m (Code E)
Runway Gradient (max)	1.25% longitudinal, 1.5% transverse
Pavement Type	Flexible (bituminous) or Rigid (PCC)
Clearway Length	150–300 m beyond runway end
Orientation	Along prevailing wind direction (wind rose analysis)
Taxiway Width	23 m (Code C), 38 m (Code E)

A.2 Wind Rose Diagram

Runway orientation is determined using a **Wind Rose Diagram** – a polar diagram showing wind speed and direction frequency. The runway is oriented along the direction with maximum wind coverage (≥95% wind coverage required by ICAO).

AIRPORT KEY FORMULA

PCN (Pavement Classification Number) - identifies load-bearing strength of airport pavement

Runway Length Correction = Basic Length × (1 + 0.07/300 × elevation + 0.01 × slope correction)

TORA (Take-Off Run Available) = Runway length declared for takeoff roll

B. Railway Engineering – Technical Details

B.1 Types of Rails Used in India

Type	Weight	Usage
60 kg/m (UIC 60)	60 kg/m	High-speed and heavy freight routes
52 kg/m (standard)	52 kg/m	Standard broad gauge routes
45 kg/m	45 kg/m	Older routes; being upgraded

B.2 Track Gauge Types

Gauge	Track Width	Used In
Broad Gauge (BG)	1676 mm (5' 6")	Main Indian Railway network (predominant)
Metre Gauge (MG)	1000 mm (3' 3⅜")	Being converted to BG under Project Unigauge
Narrow Gauge (NG)	762 mm or 610 mm	Hilly terrain – Darjeeling, Matheran
Standard Gauge	1435 mm	Metro rail systems (Delhi, Hyderabad, Mumbai)

B.3 Sleeper Types and Spacing

- **Wooden Sleepers:** Good resilience, high cost, timber scarcity – being replaced
- **Pre-Stressed Concrete (PSC) Sleepers:** Most widely used in India; durable (30–50 year life), heavy
- **Steel Sleepers:** Used in curves and special locations
- **Cast Iron Pot Sleepers:** Used in yards and sidings
- Standard Spacing: 600 mm (PSC), 550 mm (wooden) centre-to-centre

C. Harbour Engineering – Technical Details

C.1 Types of Harbours

Type	Description	Example
Natural Harbour	Protected by natural landform (bay, headland)	Visakhapatnam, Mumbai
Artificial Harbour	Created by breakwaters and dredging	Chennai, Paradip
Refuge Harbour	For shelter during storms; not a trading port	Emergency ports
Military Harbour	Naval dockyard for warships	Karwar Naval Base

C.2 Types of Breakwaters

- **Rubble Mound Breakwater:** Rock rubble dumped in water, most common type; easy to repair; slopes both sides
- **Vertical Wall Breakwater:** Vertical concrete/masonry wall; wave reflection causes scour at base
- **Composite Breakwater:** Vertical wall on rubble mound foundation
- **Floating Breakwater:** Used in moderate wave conditions; pontoon-based

▢ HARBOUR ENGINEERING KEY RELATIONS

Hudson's Formula (Breakwater design): $W = \gamma_r \cdot H^3 / (K_D \cdot (S_r - 1)^3 \cdot \cot \theta)$

where W = weight of armour unit (N), H = design wave height (m)

S_r = specific gravity of rock, K_D = stability coefficient, θ = slope angle

Tidal Range = High Water Level - Low Water Level

Ship's Draft = depth of keel below waterline; must be less than water depth in harbour

TUNNEL ENGINEERING – DETAILED COVERAGE

Tunnel Engineering – Methods of Construction

Method	Used For	Description
Drill and Blast	Rock tunnels (mountains)	Explosives used to break rock; debris cleared; lining applied
Tunnel Boring Machine (TBM)	Metro, water tunnels, soft ground	Rotating cutterhead bores continuously; most modern method
Cut and Cover	Shallow urban tunnels	Excavate from surface, build tunnel, cover back; used for metro
New Austrian Tunneling Method (NATM)	Rock mass tunnels	Utilizes rock self-supporting capacity; shotcrete + rock bolts
Immersed Tube	Under rivers/harbours	Pre-fabricated tunnel sections sunk into dredged trench

Tunnel Components

- **Portal:** Entry/exit structure of tunnel with retaining walls and drainage
- **Lining:** Concrete/shotcrete/steel segments lining the tunnel bore for structural support
- **Waterproofing Membrane:** HDPE membrane between rock and lining to prevent water ingress
- **Drainage System:** Longitudinal and transverse drains to remove seepage water
- **Ventilation System:** Jet fans or forced ventilation to maintain air quality inside
- **Emergency Exit:** Cross-passages at regular intervals for evacuation

More Q&A – Important Exam Questions

5 Marks

Q11. Explain sources of water and quality parameters of water for public supply.

SOURCES OF WATER: Water for public supply can be obtained from:

(A) Surface Sources:

(i) Rivers – largest source; perennial rivers preferred; water may be turbid and need treatment

- (ii) Lakes/Ponds – natural or artificial; quality depends on catchment land use
- (iii) Irrigation Tanks/Reservoirs – stored rainwater in dams; most dependable
- (iv) Streams – smaller watercourses; seasonal and not always reliable

(B) Sub-Surface/Ground Sources:

- (i) Open Dug Wells – 5–15 m deep; collects shallow groundwater; contamination risk
- (ii) Tube Wells/Bore Wells – 50–300 m deep; uses pumps; better quality
- (iii) Artesian Wells – groundwater under pressure; flows without pumping where confined aquifer is tapped
- (iv) Springs – where groundwater naturally surfaces; high quality

(C) Rainwater Sources:

- (i) Rooftop Rainwater Harvesting – collected through gutters and storage tanks
- (ii) Surface Collection Tanks – large open tanks to collect surface runoff

QUALITY PARAMETERS:

1. Physical Parameters:

- Turbidity (≤ 1 NTU) – cloudiness due to suspended particles
- Colour (≤ 5 Hazen units) – due to dissolved organics/minerals
- Taste and Odour – presence of H_2S , algae, or dissolved gases
- Temperature – affects chemical reactions and bacterial growth

2. Chemical Parameters:

- pH (6.5–8.5), TDS (≤ 500 mg/L), Total Hardness (≤ 200 mg/L)
- Chlorides (≤ 250 mg/L), Sulphates (≤ 200 mg/L)
- Heavy metals: Iron (≤ 0.3), Arsenic (≤ 0.01), Lead (≤ 0.01) in mg/L
- Nitrates (≤ 45 mg/L), Fluorides (≤ 1.0 mg/L)

3. Biological Parameters:

- E. coli – must be Absent in 100 mL sample
- Total Coliform – must be Absent; indicates fecal contamination
- BOD (≤ 2 mg/L for treated supply) – organic pollution indicator
- DO (≥ 4 mg/L) – necessary for aquatic life; low DO indicates pollution

Reference: *IS 10500:2012 – Indian Standard Drinking Water Specification*

2 Marks

Q12. Distinguish between a reservoir and a dam.

Difference between Dam and Reservoir:

Dam: A dam is a *structure* – a physical barrier constructed across a river using concrete, masonry, or earth materials to block and control water flow. The dam itself is the engineering structure.

Key features: Has foundations, body, crest, spillway, sluice gates, and drainage system.

Reservoir: A reservoir is the *water body* or storage body formed behind the dam when river water is impounded.

It is not a structure but a storage volume.

Key features: Has defined storage zones – MWL (Maximum Water Level), FRL (Full Reservoir Level), MDDL (Minimum Draw Down Level), and Dead Storage.

Relationship: Dam → creates → Reservoir. A dam without a reservoir has no storage function. The reservoir stores water; the dam controls its release.

Example: Bhakra Dam (concrete gravity dam) creates the Gobind Sagar Reservoir (largest reservoir in India by volume = 9.34 km³)

SECTION 17: TEXTBOOK & REFERENCE LIST



Prescribed Textbooks (JNTUK R23)

- [T1] **M.S. Palanisamy**, "Basic Civil Engineering," Tata McGraw Hill Education, New Delhi. (*Chapters on Transportation, Water Resources, and Environmental Engineering*)
- [T2] **S.S. Bhavikatti**, "Introduction to Civil Engineering," New Age International Publishers, New Delhi, 2022. (*Covers pavement types, hydrology basics, and dam structures*)
- [T3] **Satheesh Gopi**, "Basic Civil Engineering," Pearson Education, 2009. (*Comprehensive coverage of transportation and water resources for first-year students*)



Reference Books

- [R1] **S.K. Duggal**, "Surveying Vol I & II," Tata McGraw-Hill, New Delhi. (*Reference for measurement and civil engineering fundamentals*)
- [R2] **Santosh Kumar Garg**, "Hydrology and Water Resources Engineering," Khanna Publishers, New Delhi. (*Comprehensive reference for hydrological cycle, runoff analysis, and water balance*)
- [R3] **Santosh Kumar Garg**, "Irrigation Engineering and Hydraulic Structures," Khanna Publishers, New Delhi. (*Dam design, reservoir operation, and canal engineering*)
- [R4] **Khanna, Justo & Veeraraghavan**, "Highway Engineering," Nem Chand Brothers, Roorkee. (*Pavement design, geometric design, IRC codes*)
- [R5] **Bureau of Indian Standards**, "IS 10500:2012 – Indian Standard Drinking Water Specification (Second Revision)," BIS, New Delhi. (*Mandatory reference for water quality parameters*)



Indian Standards (IS Codes) Referenced

IS Code	Title	Relevance
IS 10500:2012	Drinking Water Specification	Water quality limits for all parameters
IS 456:2000	Plain and Reinforced Concrete	Concrete design for rigid pavement and dams
IS 15797:2008	Rainwater Harvesting Guidelines	Design of RWH systems
IRC:37-2018	Flexible Pavement Design	Design of flexible pavements for highways

IRC:58-2015	Rigid Pavement Design	Design of cement concrete pavements
IS 1893:2016	Earthquake Resistant Design	Seismic design of dams and structures

Glossary of Key Technical Terms

Term	Definition
CBR	California Bearing Ratio – measure of subgrade strength
TDS	Total Dissolved Solids – total mineral content of water
BOD	Biochemical Oxygen Demand – organic pollution indicator
NTU	Nephelometric Turbidity Units – measure of water cloudiness
FRL	Full Reservoir Level – design top water storage level
MDDL	Minimum Draw Down Level – lowest usable water level in reservoir
MWL	Maximum Water Level – extreme flood level in reservoir
FOS	Factor of Safety – ratio of resistance to applied load
ESAL	Equivalent Single Axle Load – standard traffic loading unit
DLC	Dry Lean Concrete – sub-base for rigid pavement (M10)
WMM	Wet Mix Macadam – base course for flexible pavement
GSB	Granular Sub-Base – granular layer below base course
PSC	Pre-Stressed Concrete sleepers for railway track
EIA	Environmental Impact Assessment – pre-project environmental study
ICAO	International Civil Aviation Organisation – aviation standards body
BRO	Border Roads Organisation – constructs strategic roads in India
TBM	Tunnel Boring Machine – used for boring tunnels mechanically

ADDITIONAL 2-MARK Q&A AND UNIT SUMMARY

2 Marks

Q13. What is tunnel engineering? State any two types of tunnels.

Tunnel Engineering: Tunnel engineering is the branch of civil engineering that deals with planning, design, and construction of underground passages through mountains, under rivers, through cities, or under seabeds. These passages are used for roads, railways, water, metro, or utilities.

The science involves soil/rock mechanics, groundwater control, ventilation design, structural lining design, and construction methodology (drilling, blasting, TBM boring, or cut and cover).

Types of Tunnels:

- (i) **Road Tunnel:** Underground passage for vehicular traffic through mountains or under rivers. Example: Zoji La Tunnel, J&K (world's longest bi-directional tunnel at 14.2 km).
- (ii) **Metro/Railway Tunnel:** Underground passage for trains/metro in urban areas. Example: Delhi Metro underground sections use TBM-bored tunnels at 10–20 m depth.
- (iii) **Water Tunnel:** Underground conduit for conveying water for hydro-power (penstocks) or irrigation. Example: Tehri Hydro Project water tunnels.
- (iv) **Cable/Utility Tunnel:** Carries electricity cables, water pipes, telecom cables underground in urban areas.

2 Marks

Q14. What is environmental engineering? State its importance.

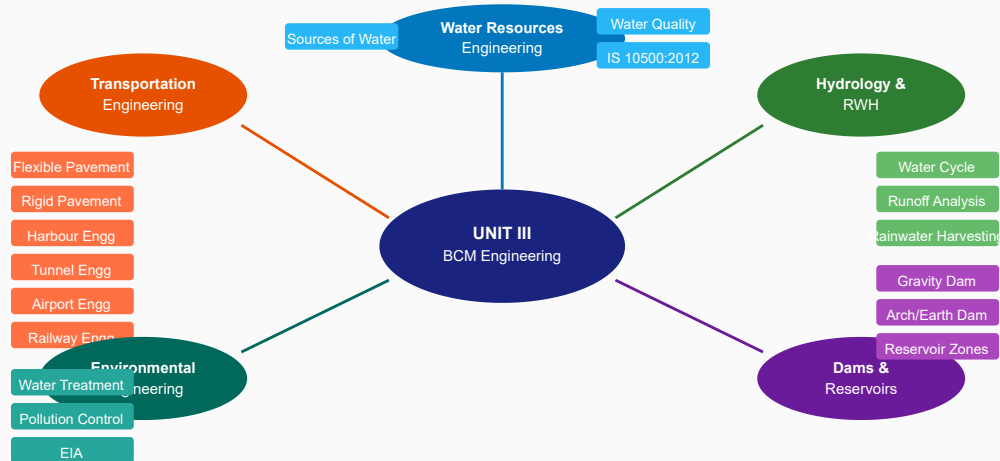
Environmental Engineering: Environmental Engineering is the branch of engineering that applies scientific and engineering principles for environmental protection and improvement. It deals with water supply, sewage treatment, solid waste management, air pollution control, and sustainable development.

Importance of Environmental Engineering:

- (i) **Public Health:** Safe water supply and sewage disposal prevent waterborne diseases (cholera, typhoid, dysentery). WHO estimates 80% of diseases in developing countries are water-related.
- (ii) **Environmental Protection:** Prevents water bodies from becoming polluted by industrial and domestic effluents, protecting aquatic ecosystems.
- (iii) **Sustainable Development:** Balances industrial development with environmental preservation through EIA, waste minimization, and green technology.
- (iv) **Climate Adaptation:** Designs resilient infrastructure for flood control, drought management, and urban heat island mitigation.
- (v) **Regulatory Compliance:** Ensures industries comply with pollution control norms under Environment Protection Act 1986 and CPCB guidelines.



Fig: Unit III Complete Mind Map



JNTUK R23 | Basic Civil & Mechanical Engineering | Unit III

✦ Final Exam Strategy for Unit III:

- ✓ Draw all diagrams neatly with labels – partial marks are given for diagrams even if theory is incomplete.
- ✓ For comparison questions, always use a table format with minimum 5 points.
- ✓ Mention IS code numbers wherever applicable – shows depth of knowledge.
- ✓ For derivations/formulas, define all variables clearly.
- ✓ Unit III carries 20–25% weight in JNTUK End Semester Exam (14–20 marks out of 70).

UNIT III – COMPLETE

Basic Civil & Mechanical Engineering | JNTUK R23 | I Year I Semester

Transportation • Water Resources • Hydrology • Dams • Environmental Engineering

References: Palanisamy | Bhavikatti | Satheesh Gopi | IRC:37 | IS 10500 | Santosh Kumar Garg